

Motivation

- Recent Studies (Patricola et al., 2018; Danso et al. 2022) have shown that African Easterly Waves don't control the number of TCs in the Atlantic.
- The ITCZ in the Eastern North Pacific (ENP) is highly active during summer and fall, and it can undergo breakdown either by itself or with Tropical Easterly Waves (TEWs). (Wang & Magnusdottir, 2006).
- Large-scale environmental parameters, such as the Genesis Potential Index (GPI), are not effective in predicting storm hotspots in the ENP, regardless of the reanalysis dataset or the specific formulation of the GPI used (Sobel et al., 2021; Chavas et al., 2024*).

Research questions

- Does seed structure vary significantly based on the disturbance that produces it?
- Does ITCZ related seed contribute significantly to the number of TCs in the ENP?

Data and Methods

For seeds we use the database from Ikehata and Satoh, 2021.

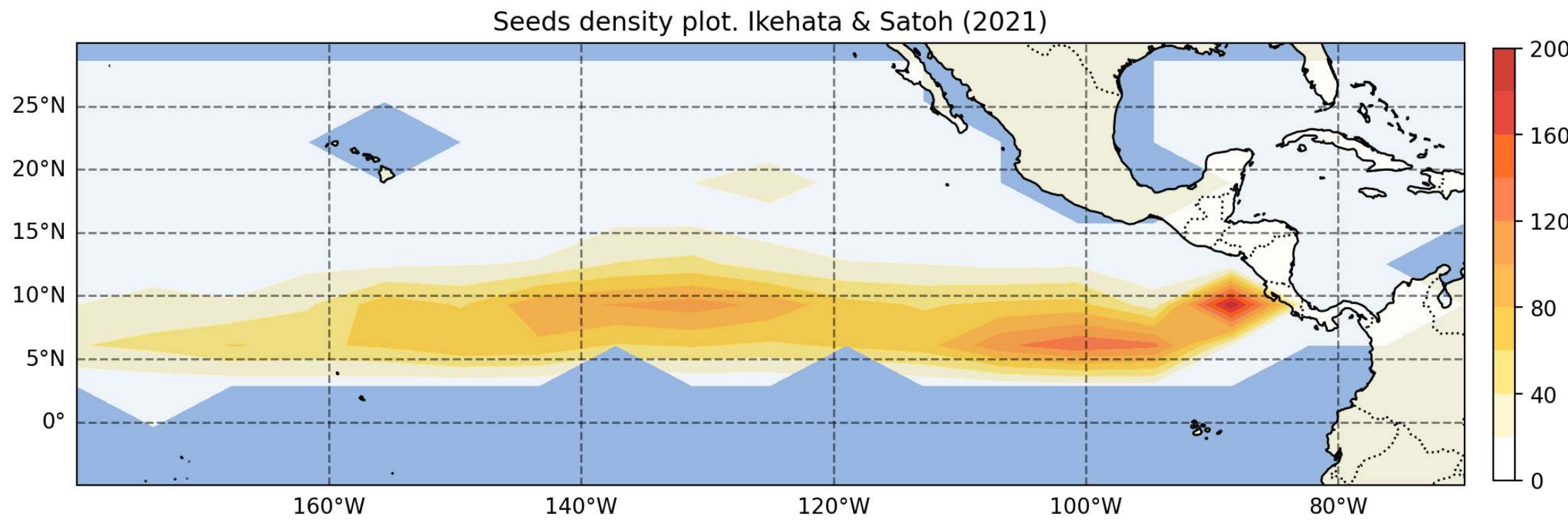


Fig 1. Total number of seeds in database from Ikehata & Satoh (2021).

For TEWs we use the database from Hollis et al., 2024.

- Shallow TEWs are waves only present on the 700hPa dataset and deep are collocated waves that appear on both the 700 and 850hPa.

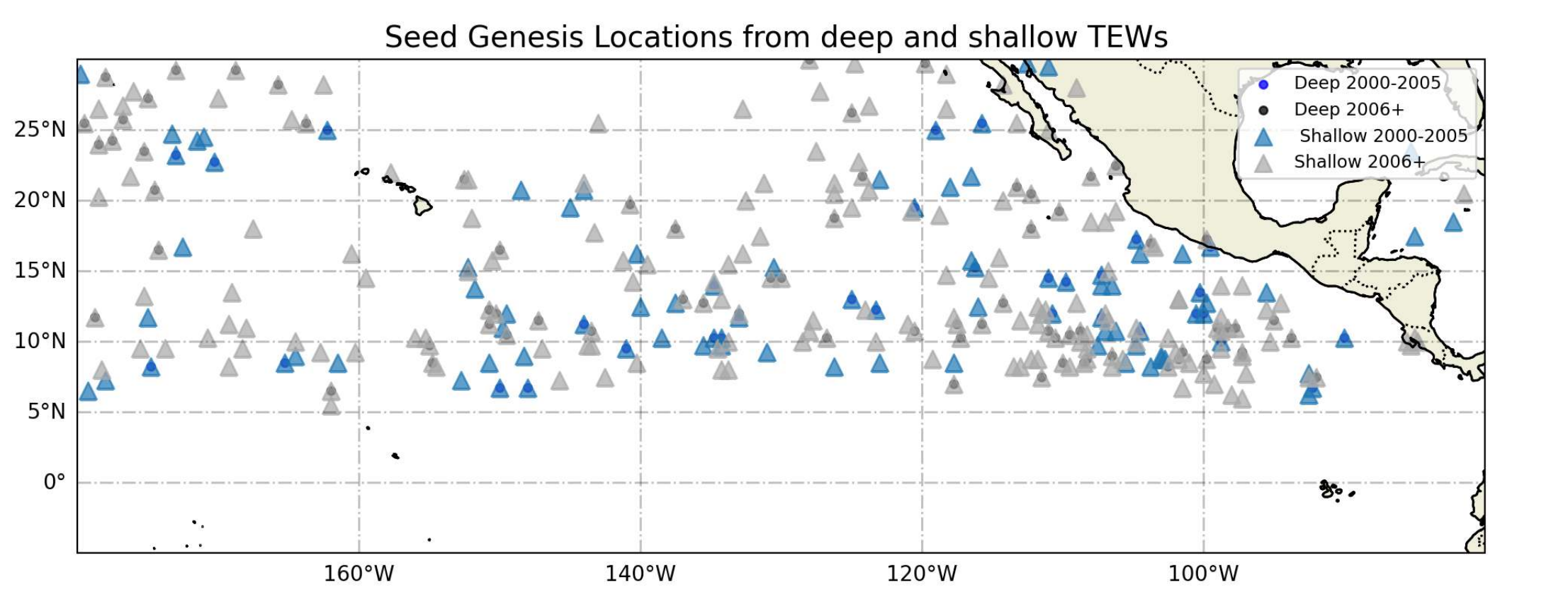


Fig 2. Seeds genesis points from deep and shallow TEWs. A seed is related to a TEW if its genesis points is within 500km of the any wave track during the same period.

For purely ITCZ breakdowns we use only the breakdowns form by the “vortex rollup” mechanism from table 3 of Wang and Magnusdottir, 2006.

For locating the ITCZ breakdowns we use ERA5 data as follows:

- Breakdowns form from ITCZ bands: a band of at least 4° long with minimum value of $0.5 \times 10^{-4} \text{ s}^{-1}$ of 975hPa relative vorticity.
- At least half of the points within the ITCZ band are in a region of high (top 5 percentile) convergence at 975hPa.
- Using instantaneous OLR classify the region as deep or shallow convection.

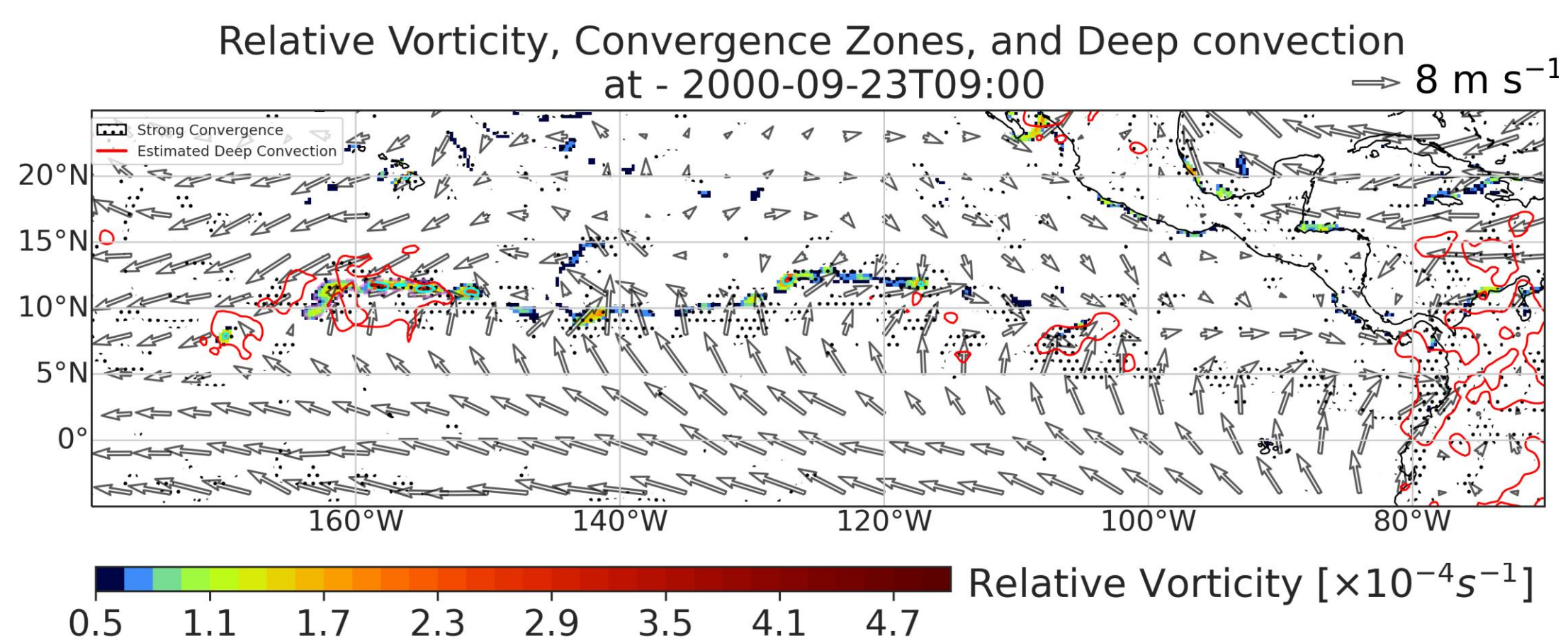


Fig 3. Example of ITCZ breakdowns detection in the ENP. Purple dashed contours indicate ITCZ bands that could undergo breakdown and cyan countours are regions with Relative vorticity $\geq 3 \times 10^{-4} \text{ s}^{-1}$

Results 1: The rarer "deeper" TEWs produce stronger seeds in better environments.

- Seeds from deep TEWs spin faster.
- Seeds from deep TEWs are in a moister environment, not only in the ascendent region.

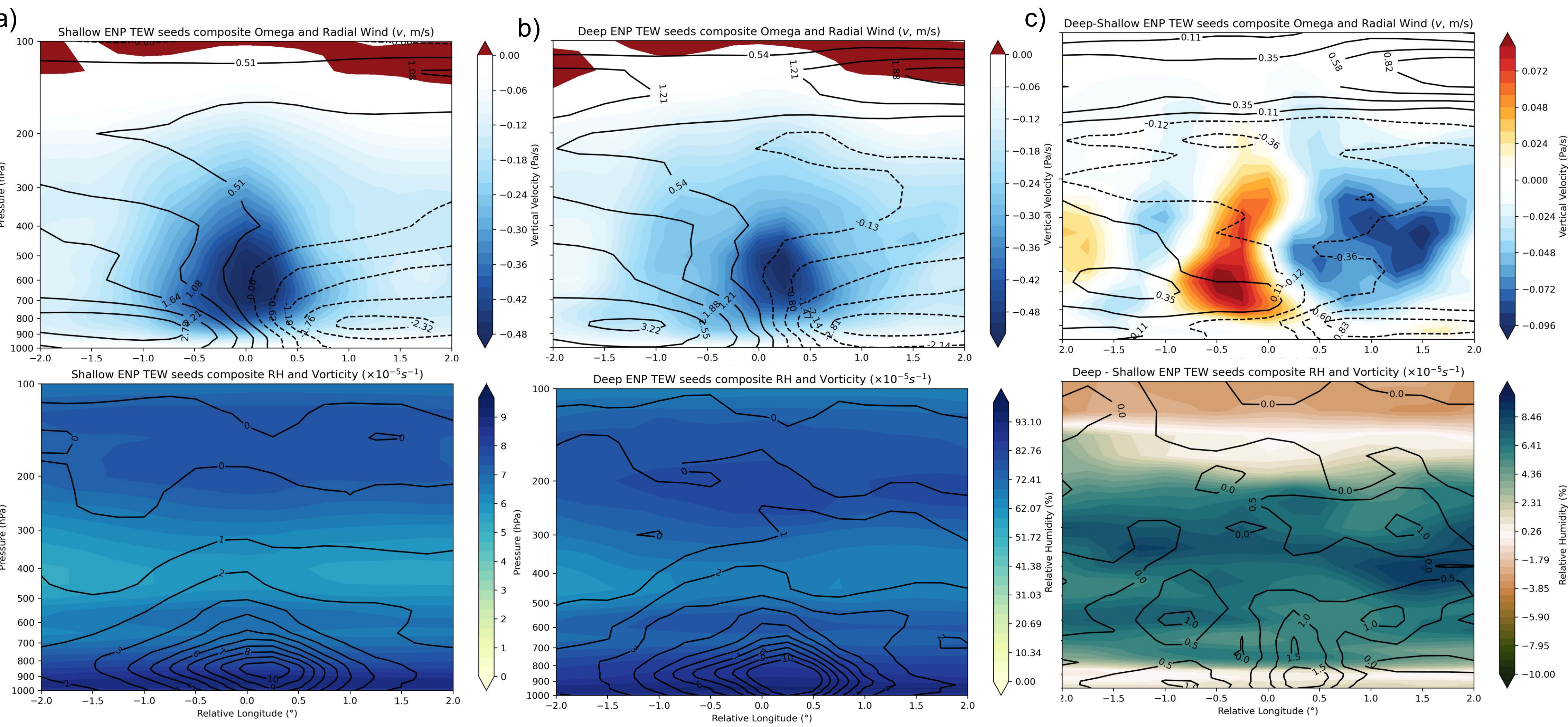


Fig 4. Vertical cross sections of seeds coming from a) shallow, b) deep TEWs, and c) their difference. The first row shows the vertical velocity in shadings and contours of radial velocity. The second row shows relative humidity in shadings and contours of relative vorticity.

Results 2*: ITCZ Breakdowns seeds are deeper than TEWs seeds

- Seeds from ITCZ breakdowns spin faster than both seeds from TEWs.
- Seeds from ITCZ has a moister lower troposphere but drier middle troposphere.

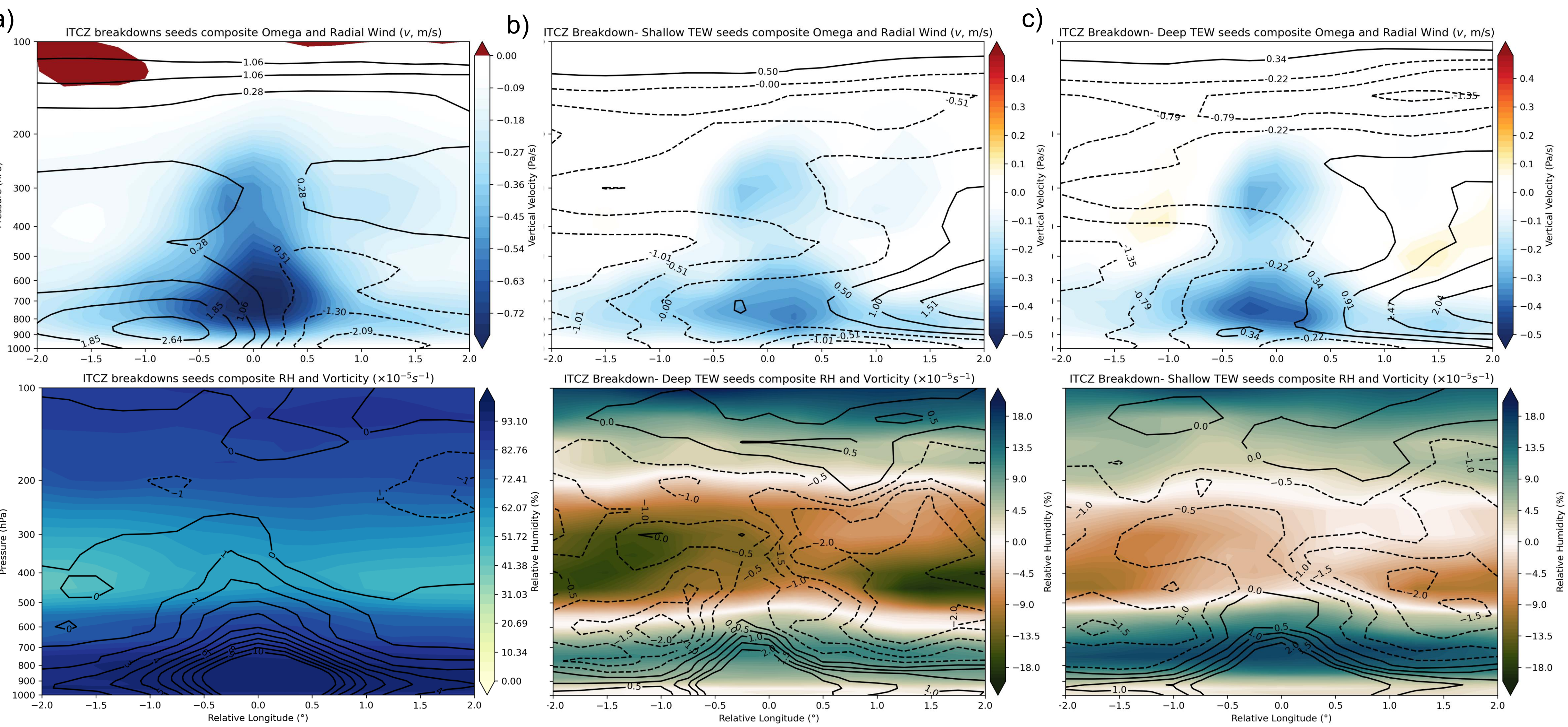


Fig 5. Vertical cross sections of seeds coming from a) ITCZ breakdowns, and their differences with b) shallow, and c) deep TEWs. Each row showing the same variables as in Fig3.

Ongoing and Future Work

- Currently automatizing the ITCZ breakdown algorithm, including a check on incorporating barotropic instability criteria (Rayleigh-Kuo + Fjortoft) to extend the analysis on seeds from breakdowns to the satellite era.
- To test the importance and effects of TEWs seeds on the Eastern North Pacific we are running simulations on MPAS for 3 seasons (highly, lower, and average TC activity) in the Eastern Pacific with and without TEWs.
- To test the importance and effects of ITCZ breakdown seeds we are running high resolution (convective-permitting) and low resolution (according to Wang and Magnusdottir, 2005 minimum resolution for capturing breakdown is 1°) on the same 3 season for the ENP.

# TEWs Deep seeds	106
# TEWs Shallow seeds	286
# ITCZ breakdown seeds	44

Let's connect ☺

